GeoFlux

Visualizing 30 Years of Global Commodity Trade

An interactive data visualisation dashboard exploring world trade flows from 1988 to 2016, across 112 countries, 16 trade categories and hundreds of commodities. Built with Nuxt, Mapbox GL and D3.js.

NUXT 4 · MAPBOX GL JS · D3.JS · TYPESCRIPT · UN COMTRADE · KAGGLE TRADE DB · TAILWIND CSS V4

TEAM MEMBERS

Coralie Banuls Jason Miller Jaime Lopez Lucas Martiniano

COURSE
COM-480 Data Visualisation

INSTITUTION
EPFL — Spring 2026

DATA PERIOD
1988 to 2016

INTRODUCTION

Why GeoFlux?

Over the three decades covered by our dataset, the world experienced profound macroeconomic shifts: the fall of the Soviet Union, the rise of China as a global manufacturing superpower, the dot-com boom, and the 2008 financial crisis. All of these events left fingerprints in trade data. GeoFlux was born from one question — can we make those fingerprints visible and explorable by anyone?

DATASETS

Two Complementary Sources

PRIMARY **UN Comtrade — Global Commodity Trade Statistics**

Published by the United Nations Statistics Division and available on Kaggle. The dataset contains **8,225,871 records** and 10 variables, covering international trade between **209 countries** from 1988 to 2016. Key fields include country, commodity category, trade direction (import or export), trade value in USD, and weight in kg.

This dataset powers every page: the globe choropleth, country detail windows, category and commodity pages, evolution charts, stat cards and supply barometers.

SECONDARY **Kaggle — International Trade Database**

A second dataset was required for the **Trade Flows** feature. The primary UN dataset records what each country trades and how much, but not *with whom*. The Kaggle International Trade Database fills this gap with bilateral partner data, enabling us to compute the top 3 import sources and export destinations per country per year.

Limitation: the secondary dataset does not carry sufficient category-level breakdowns. Trade flow arcs on the map therefore represent total trade between country pairs and cannot be filtered by category.

8.2M TRADE RECORDS

112 COUNTRIES

29 YEARS COVERED

90+ CATEGORIES

AUDIENCE AND OBJECTIVES

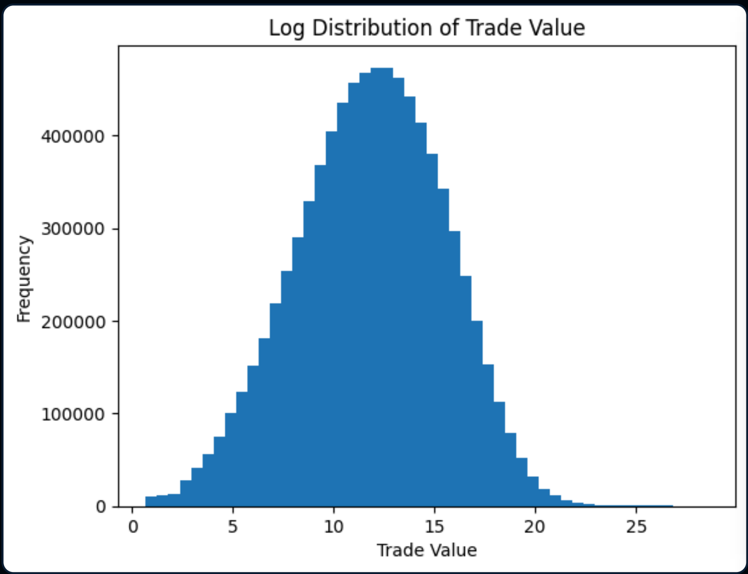
GeoFlux targets students, economics enthusiasts, and anyone curious about geopolitics. No prior knowledge of trade statistics is required. The interface rewards exploration at any depth — from a quick choropleth glance to drilling from a country through its categories down to individual commodities.

- Explore how China's trade dominance grew from 2000 onward
- Compare import and export intensities in any year
- Drill from globe to country to category to commodity
- Visualise bilateral trade connections as animated arcs
- Filter the entire map by trade category

MILESTONE 1 — EXPLORATORY DATA ANALYSIS

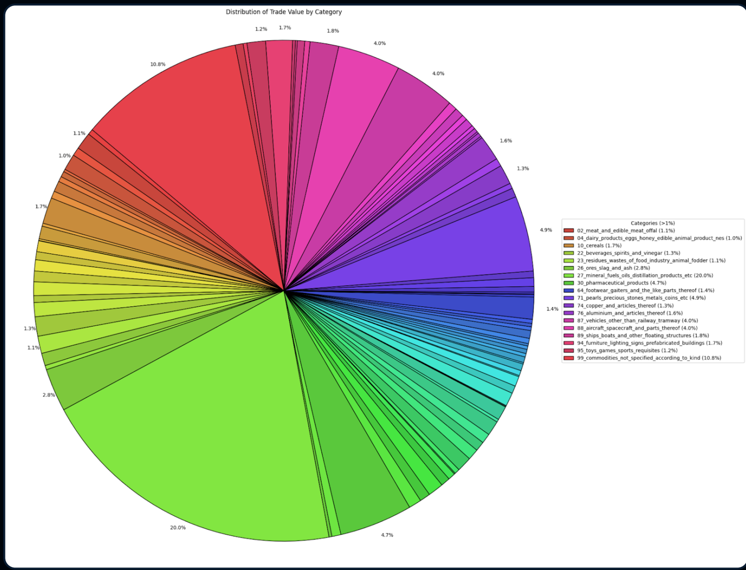
Understanding the Data Before Designing

Before any design decision was made, we explored the raw dataset in Python. Four findings shaped design choices that remained stable throughout the entire project.



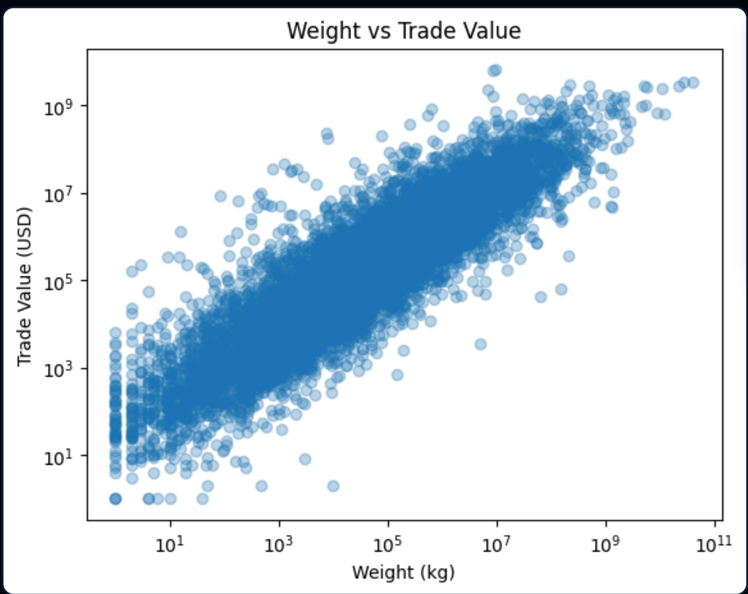
FINDING 1 — EXTREME VALUE SKEW

Trade values span many orders of magnitude. A linear scale makes most countries invisible. This led us to adopt a **power scale (exponent 0.35)** for the choropleth, compressing the China and US dominance while preserving gradient readability for smaller economies.



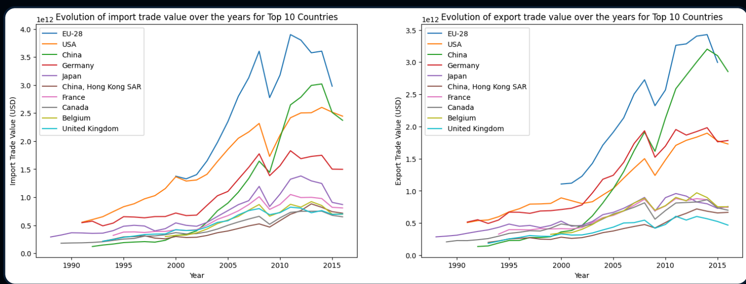
FINDING 3 — CATEGORY CONCENTRATION

The top 5 categories (Mineral Fuels, Precious Stones, Pharmaceuticals, Vehicles, Aircraft) account for roughly 43% of total world trade value. This justified building dedicated **category and commodity detail pages** rather than keeping the data buried in country pop-ups.



FINDING 2 — WEIGHT VS. VALUE SPLIT

Raw materials are heavy and cheap; electronics and pharmaceuticals are light and expensive. This structural split directly motivated the **USD / Weight toggle** that runs through the entire dashboard — switching it reveals a completely different geographical distribution.



FINDING 4 — CHINA'S RISE

China overtook most economies in export volume from 2000 onward, exhibiting uniquely rapid growth driven by WTO accession. This story arc became the centrepiece of the **Trade Evolution** race chart and motivated the year slider spanning all 29 years of the dataset.

MILESTONE 2 — DESIGN PLAN AND FIRST SKETCHES

The Three-Page Architecture

By Milestone 2 we had crystallised a three-page structure. Each sketch represented a distinct interactive surface, with the globe as the persistent entry point across all of them.

PAGE 1 — COUNTRY POP-UP

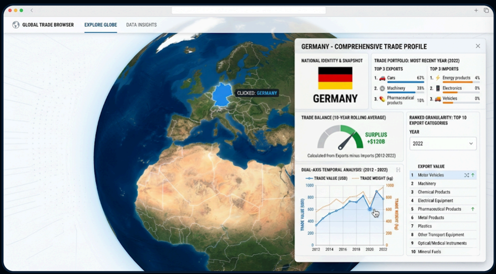
Clicking a country opens a dashboard: top 3 imports and exports for the most recent year, trade balance as a 10-year rolling average, dual-axis temporal charts, top 10 export categories.

PAGE 2 — COMMODITY EXPORT MAP

The globe dynamically recolours nations based on user-selected criteria: commodity category, year, and metric (value or weight). A left panel hosts all the filter controls.

PAGE 3 — YEAR DASHBOARD

A big horizontal year slider drives all charts simultaneously: top 10 exporters by weight and value, top 10 commodities, a treemap of category market share, pop-up notes for historical events.



Sketch — Globe with country pop-up dashboard

Sketch — Commodity export choropleth

Sketch — Data analysis dashboard over time

MILESTONE 2 — MVP AT SUBMISSION

MVP — Interactive world map with basic coloring

MVP — Early country pop-up panel

FROM PLAN TO FINAL — WHAT CHANGED AND WHY

Architecture Evolution

MILESTONE 2 PLAN		FINAL IMPLEMENTATION
Separate pages for filters (Page 2) and pop-up (Page 1)	→	Single persistent map with floating left-panel control bar
Country drill-down only — no sub-levels beyond categories list	→	Four-level drill-down: globe → country → category → commodity
Year dashboard as a standalone page with treemap	→	Year window with grouped bar charts and historical context note
Trade flows not planned	→	Animated bilateral arc lines, colour-coded per flow direction
No world-level overview page	→	Global Statistics window: bar charts, categories pie, evolution race
No search functionality	→	Full-text search across countries, categories, commodities and years
Globe only	→	Globe and flat Equal Earth projection toggle for whole-world view

FINAL GLOBE — THREE CHOROPLETH STATES

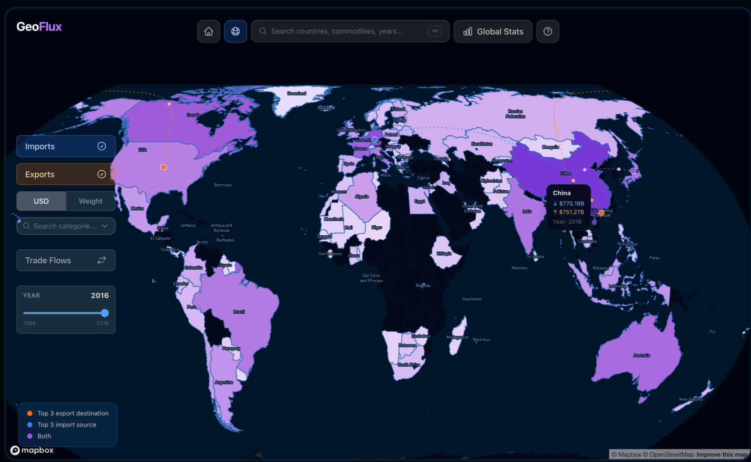
Imports active — blue scale

Exports active — orange scale

Both active — purple scale

EQUAL EARTH FLAT PROJECTION — WHOLE-WORLD OVERVIEW

Flat view, Vehicles category filter, year 1999



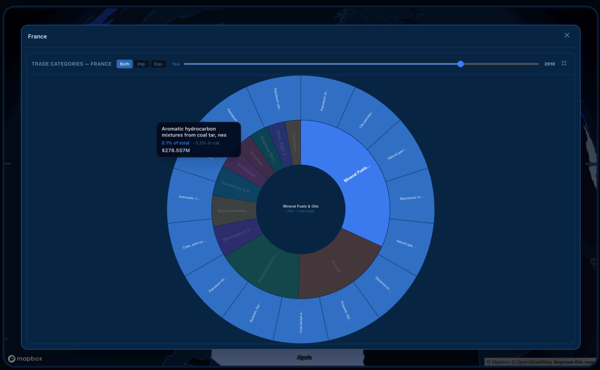
Flat view, imports and exports combined, China tooltip

DETAIL PAGES — ADDED BEYOND THE ORIGINAL PLAN

Four Levels of Exploration

The most significant addition to our Milestone 2 plan was the creation of dedicated pages for categories and commodities. Every element across every chart is navigable — clicking a pie slice, a bar label, or a country name pushes a new window onto the breadcrumb stack.

Country page — France, year 2010



Expanded sunburst — France trade categories breakdown

Category page — Mineral Fuels and Oils, year 1991

COUNTRY PAGE

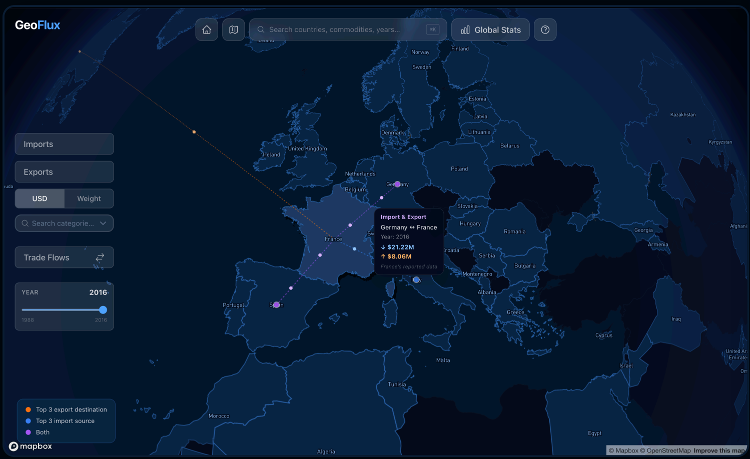
- Year slider synced with the global map
- 5 stat cards: imports, exports, balance, 10-year trend, world share
- Top 3 import sources and export destinations, each clickable
- Category pie chart — expandable to full SunburstPie
- Evolution race chart, autoplay on open

CATEGORY PAGE

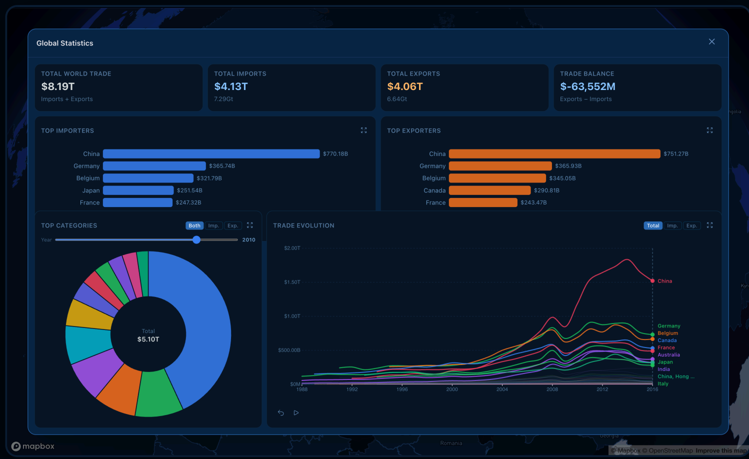
- Supply Barometer: green, amber or red based on how many of 112 countries export this category
- Commodity breakdown pie chart, year-aware proportions
- Top 5 importers and exporters as bar charts
- Evolution race chart, autoplay on open

TRADE FLOWS, GLOBAL STATISTICS, SEARCH AND YEAR PAGES

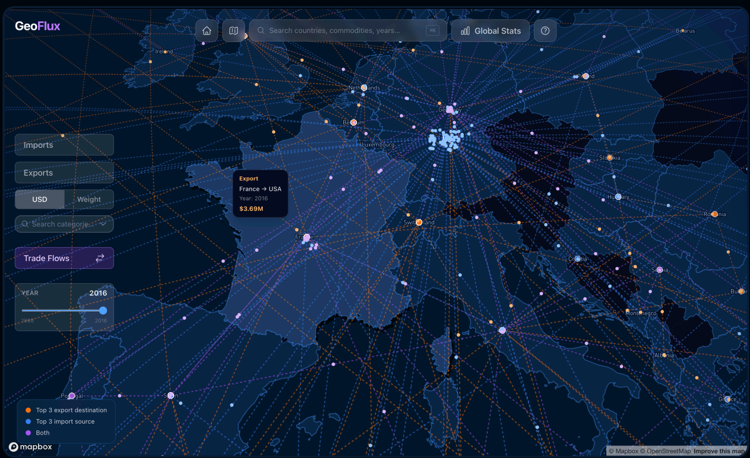
Features Beyond the Original Scope



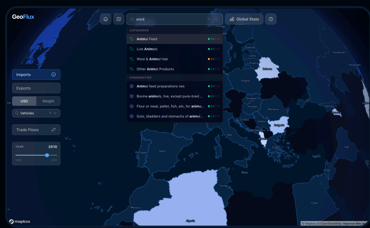
Bilateral flow hover — Germany and France connection, year 2016



Global Statistics window — bar charts, categories pie, evolution race



All trade flows active — animated arcs across Europe



Search with supply barometer badges

Year 2002 snapshot with historical context

TRADE FLOWS — TECHNICAL NOTE

Flow lines required the secondary Kaggle dataset to obtain partner-level data absent in the primary UN source. An invisible 18px hit-target layer sits above each visible 1.5px arc to allow comfortable hover detection. Because the secondary dataset lacks category breakdowns, flow arcs cannot be filtered by commodity category.

GLOBAL STATISTICS

Not in the original plan, the Global Statistics window emerged from the need for a world-level overview before drilling into countries. It features top 5 importers and exporters as horizontal bar charts, expandable to all 112 countries, a year-aware categories pie chart with import, export and both toggle, and the evolution race chart.

YEAR PAGES AND SEARCH

Each year from 1988 to 2016 has a dedicated window with global totals, a historical context note, and grouped bar charts for top categories and top countries. The search bar provides full-text lookup across all entities and shows supply barometer badges next to each category and commodity result.

KEY CHALLENGES AND SOLUTIONS

DATA SCALE AND AGGREGATION

The raw dataset has 8.2 million rows, too large for any browser. We pre-aggregated aggressively in Python: country-by-year totals, category-by-year global totals, commodity-by-year totals, and a bilateral trade connections JSON. The frontend reads only these summaries, keeping the app fully static and fast.

YEAR-SPECIFIC CATEGORY DATA PER COUNTRY

The per-category breakdown per country only exists for the 2016 reference year. For year-aware pie charts, we scale each 2016 value by the ratio of the global category total at the target year versus 2016, capturing inter-category growth rate differences so pie proportions change meaningfully across years.

BILATERAL TRADE ASYMMETRY

France's reported trade with Germany differs from Germany's reported trade with France — each country reports independently. Rather than reconciling the data, we surface this fact in the tooltip footnote. The asymmetry is itself an interesting feature of trade statistics.

STATE PERSISTENCE ACROSS NAVIGATION

All shared state — imports and exports toggles, metric, category filter, year, fly-to requests, and projection mode — lives in Nuxt useState composables persisting across client-side navigation. The map in the layout is never destroyed as users open and close country or category pages.

PEER ASSESSMENT

Coralie Banuls

CORE ARCHITECTURE, DATA AND D3

Set up Mapbox and the core dashboard architecture. Wrote the majority of D3 components (BarChartRace, PieChart, SunburstPie, GroupedBarChart, BarChart, LineChart). Handled all data processing in Python and JSON generation. Implemented the PageWindow system, navigation history, category and commodity pages, global stats, supply barometer, search enhancements, flat map toggle and GeoFlux logo. Produced the project screencast. Wrote and edited the process book.

Lucas Martiniano

MAP COLORING AND BASE CHARTS

Implemented global map coloring — choropleth for imports, exports and combined views. Disabled hover and click interactions for countries absent from the dataset. Built the initial D3 pie chart and bar chart components that are reused across multiple pages throughout the dashboard.

Jaime Lopez

CHOROPLETH EXTENSIONS AND TRADE CONNECTIONS

Expanded the choropleth to support coloring by USD, weight and individual categories, broken down by imports and exports. Collaborated on the trade connection arc rendering, animated dot system and Mapbox layer setup for bilateral flows. Contributed to the process book.

Jason Miller

SEARCH AND DOCUMENTATION

Built the full-text search functionality across countries, categories, commodities and years, including the keyboard shortcut. Collaborated on the trade connection line tooltip and interaction system. Documented the project design and architecture. Contributed to the process book.

CONCLUSION

The gap between our Milestone 2 sketches and the final result is significant, but every divergence was driven by a real user-experience motivation rather than scope creep. Adding category and commodity pages turned a two-level exploration (world to country) into a full four-level drill-down, which we believe is the most distinctive aspect of GeoFlux.

The consistent semantic colour language — blue for imports, orange for exports, purple for both — runs through every chart, map layer, button state and tooltip across the dashboard, giving the interface a coherence that makes it intuitive without needing to read labels.

EPFL COM-480 Data Visualisation, Spring 2026